one oxygen molecule to yield 2 molecules of water. The flow of electrons through the complexes spanning I to IV results in pumping of protons from the matrix across the inner mitochondrial membrane into the intermembrane space. The proton-motive force that is generated powers ATP synthesis. Complex V catalyzes ATP synthesis.

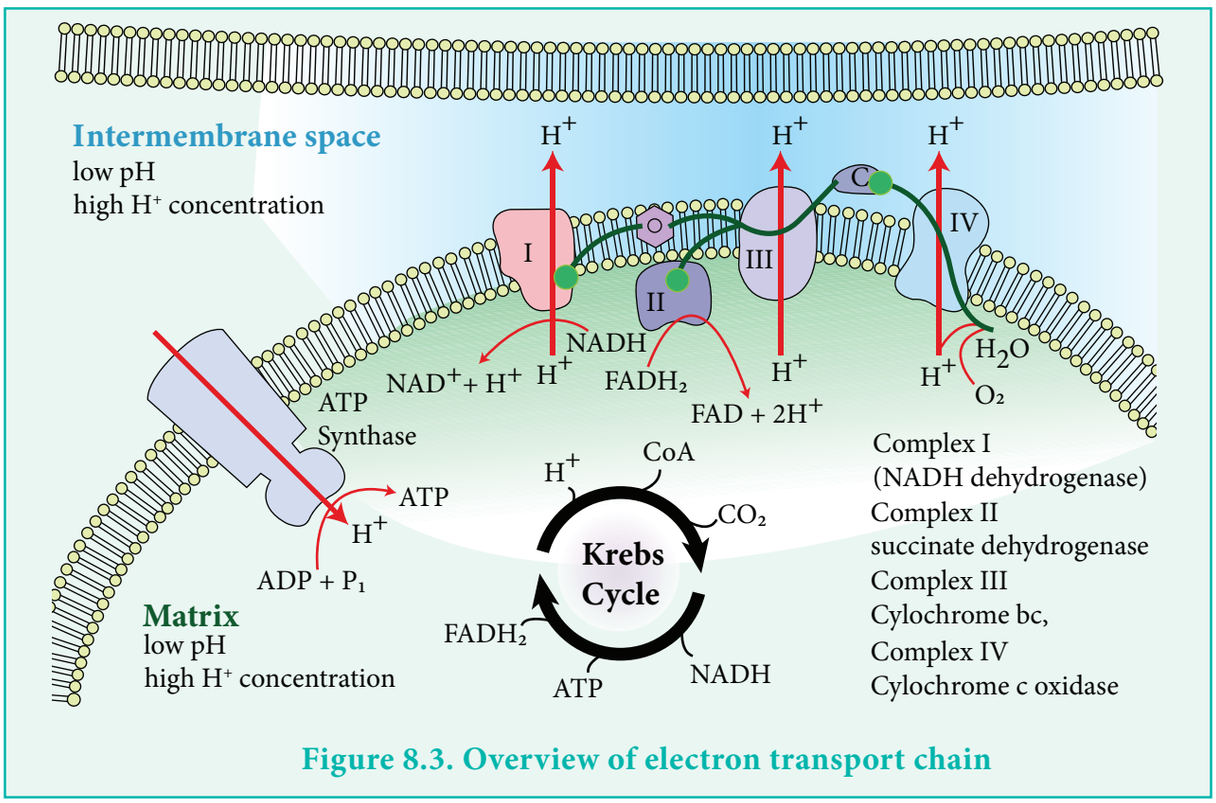


Figure 8.3. Overview of electron transport chain

A pair of electrons flowing from Complex I to Complex IV yields 3ATP, while a pair of electrons moving from Complex II to Complex IV yields 2ATP.

8.2.3 Reactions of Electron transport chain

Except coenzyme Q, all members of the respiratory chain are proteins. These proteins function as enzymes, for example, dehydrogenases. They may contain iron as part of an iron-sulfur center. Also, they may contain iron co-ordinated to a porphyrin ring as in the cytochromes, or they may contain copper, as in cytochrome a + a3 complex.

Formation of NADH: NAD^+ is reduced to NADH by dehydrogenases. which remove two hydrogen atoms from the substrate. The major sources of NADH include several reactions of the citric acid cycle, fatty acid oxidation etc. Both electrons and one proton are transferred to the NAD^+ forming NADH and a free proton H^+.

NADH dehydrogenase: The NADH and H^+ formed are transferred to NADH dehydrogenase. This complex I, is the large protein complex comprises of 46 polypeptides. is burried in the mitochondrial inner membrane and has a bound molecule of flavin mononucleotide (FMN). NADH dehydrogenase reduces FMN to $FMNH_2$. The sequential transfer of two electrons to the iron sulfur center composed of iron atoms complexed with an equal number of sulfide ions releases four protons into the intermembrane space (Fig.8.4).

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